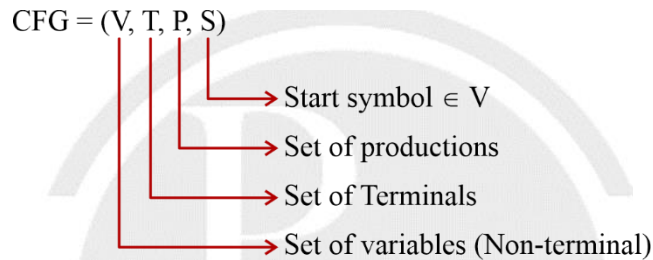


3

PUSH DOWN AUTOMATA

3.1 Context Free Grammar

CFG: It represents a Context Free Language.



- Rule of each production in P:
 $V \rightarrow (V \cup T)^*$
 V = only 1 variable in LHS
 - To derive a string, following derivations can be used.
1. **Linear Derivation:** Linear derivation is two types
 - (a) Left Most Derivation (LMD)
 - (b) Right Most Derivation (RMD)
 2. **Non- linear Derivation:** Non- linear Derivation OR Parse Tree OR Derivation Tree

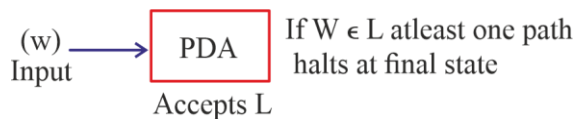
3.2 Types of Context Free Grammars

There are two types of CFG:

1. **Ambiguous CFG:** At least one string has more than one derivation.
2. **Unambiguous CFG:** Every string (w) generated by CFG has exactly one derivation.

3.3 Pushdown Automata (PDA)

PDA accepts context free language (CFL). PDA also called as NPDA.



3.4 PDA acceptance mechanisms

- PDA acceptance mechanisms are three types:
 - PDA acceptance using final state.
 - PDA acceptance using empty stack.
 - PDA acceptance using both final state and Empty stack.
 All PDA acceptance mechanisms are equivalent.
- DPDA acceptance mechanism are two types:
 - DPDA using final stack.
 - DPDA using both final state and empty stack.

3.5 PDA configuration

PDA = $(Q, \Sigma, \delta, q_0, F, Z_0, \Gamma)$.

where Q = Set of states

Σ = input alphabet

δ = Transition Function (PDA/NPDA $\delta : Q \times \Sigma \times \Gamma \rightarrow 2^{Q \times \Gamma^*}$)

F = Set of Final state

Z_0 = Bottom symbol or initially TOS

Γ = Stack Alphabet

- DPDA transition Function is $[\delta : Q \times \Sigma \times \Gamma \rightarrow Q \times \Gamma^*]$
- Difference between DPDA and PDA

DPDA	PDA
[1] $\delta : Q \times \Sigma \times \Gamma \rightarrow Q \times \Gamma^*$	$\delta : Q \times \Sigma \times \Gamma \rightarrow 2^{Q \times \Gamma^*}$
[2] Every DPDA is PDA	Every PDA need not be DPDA
[3] DPDA acceptance with - Final state mechanism - Both Final and empty stack	PDA acceptance with - Final state mechanism - Empty stack mechanism - Both Empty stack and Final state

- Relation between Regular, DCFL and CFL.

Regular languages(FA)
 DCFLs (DPDA)
 CFLs (PDA or NPDA)